

Little Explorers · Week 4

Traffic Light Controller

Let's help cars and people stay safe!

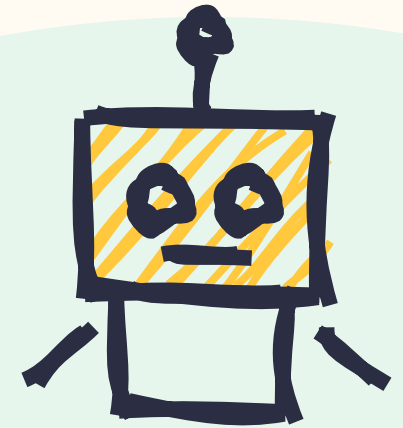
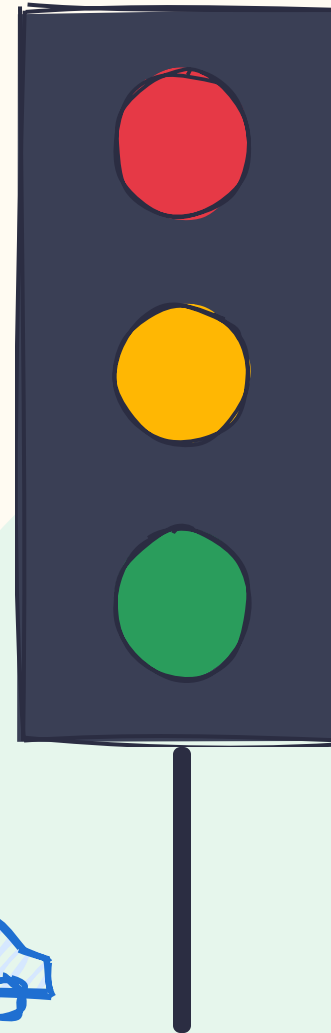
Ages 5-7

Arduino + picture blocks

60 minutes

Did you know?

The first electric traffic light was switched on in 1914, more than 100 years ago!



Let's think...

Uh-oh! The traffic lights stopped working!

Cars are coming from every side.
What will happen now?

Who goes first?

Can I cross safely?

Will cars crash?

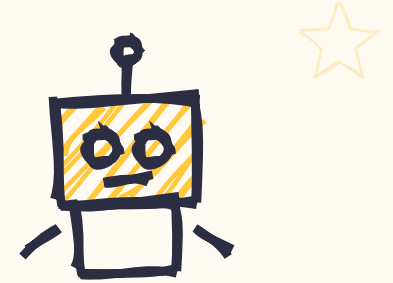
In real life

When lights break, a traffic police officer stands in the road and tells cars when to stop and go.

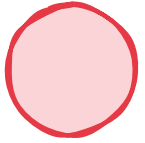


Our learning aim

Today, I will learn to...



1



I can tell what **red**, **yellow** and **green** mean.

2



I can put the lights in the **right order**.

3



I can make a real **robot brain (Arduino)** run my traffic light.

In real life

Engineers program real traffic lights in every city so millions of people stay safe every day.



Watch and think

Video time!

STEMRocket



How Do Traffic Lights Work?

How Do Traffic Lights Work?

STEMRocket · about 2 minutes · youtu.be/SWJP-RXOk6M

Watch for...

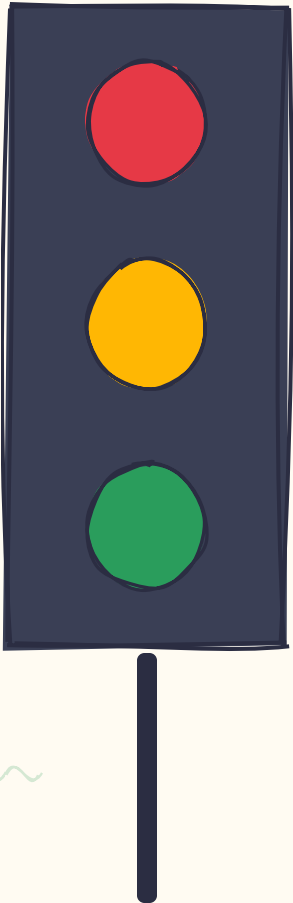
- 1 What does **red** mean?
- 2 What does **green** mean?
- 3 Who tells the lights when to change?



After the video: Play 'traffic lights' with your partner: red = freeze, amber = get ready, green = walk!

Look closely

Meet the traffic light



RED
STOP!



YELLOW
Get ready / slow down



GREEN
GO!



In real life

Look for traffic lights on your way home today. Which colour did you see?



Not just for cars

Traffic lights help people too!



Red person

Stand and wait



Green person

Now you can walk

Safety rule

Always cross on the green person and hold a grown-up's hand.



Look around you

Lights that talk to us are everywhere!



Railway crossings

Red lights flash when a train is coming



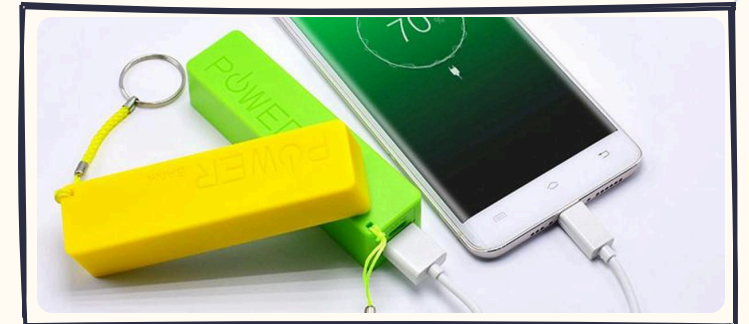
Airports

Runway lights guide planes to land safely



Chargers at home

Red = charging, green = full



Stand up and play!

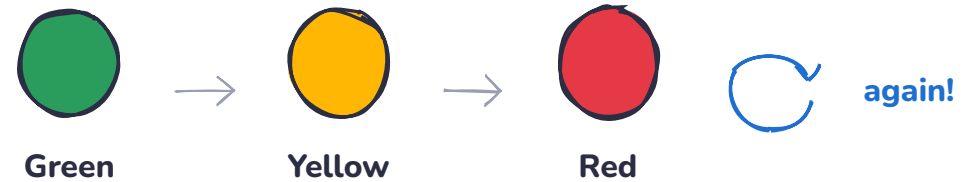
Red Light, Green Light game



GREEN: walk on the spot
YELLOW: walk very slowly
RED: freeze like a statue!

Your teacher shows the colours **in order**. Can you move the right way every time?

The order is called a sequence



In real life

Real traffic lights repeat the same order all day and all night. Repeating is called a loop.



Our builder's kit

Meet our tools



Arduino

The robot brain that follows our instructions



3 LEDs

Tiny lights: red, yellow and green



Wires

Roads for electricity to travel on

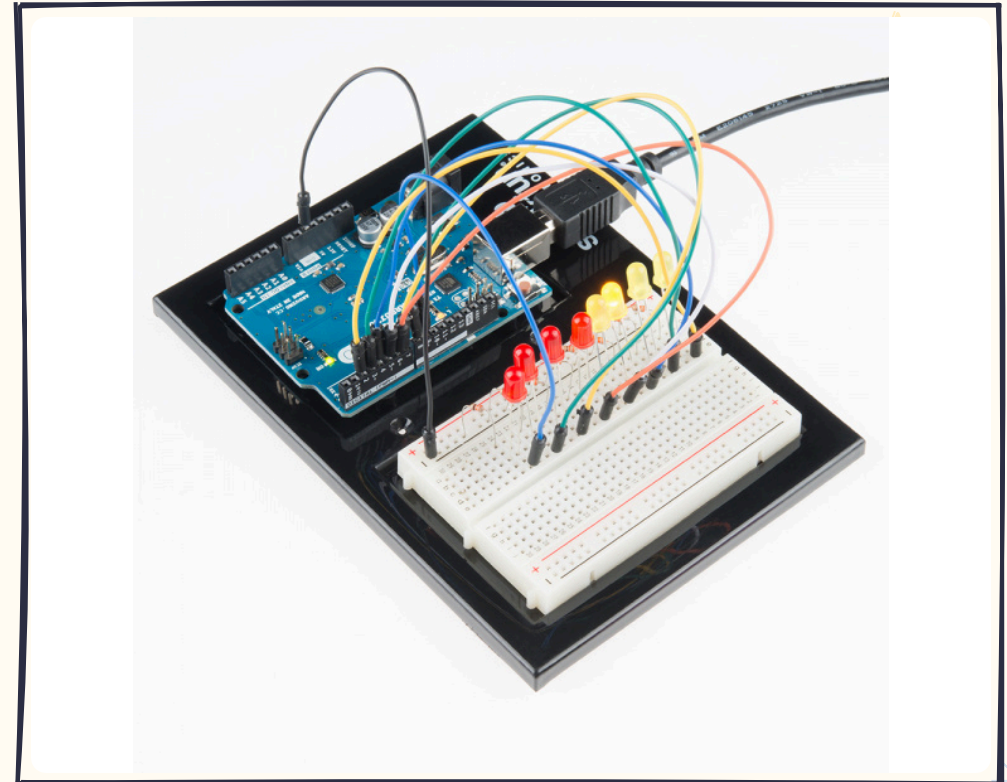


Laptop

Where we snap our picture blocks

In real life

A tiny computer like our Arduino is also inside real traffic lights, washing machines and toys!



Teacher: the circuit is already wired. Children only plug in the USB cable and program.



Time to build!

Let's code our traffic light

when Arduino starts

forever

○ turn ON green light

wait 3 seconds

○ turn OFF green light

○ turn ON yellow light

wait 1 second

○ turn OFF yellow light

● turn ON red light

wait 3 seconds

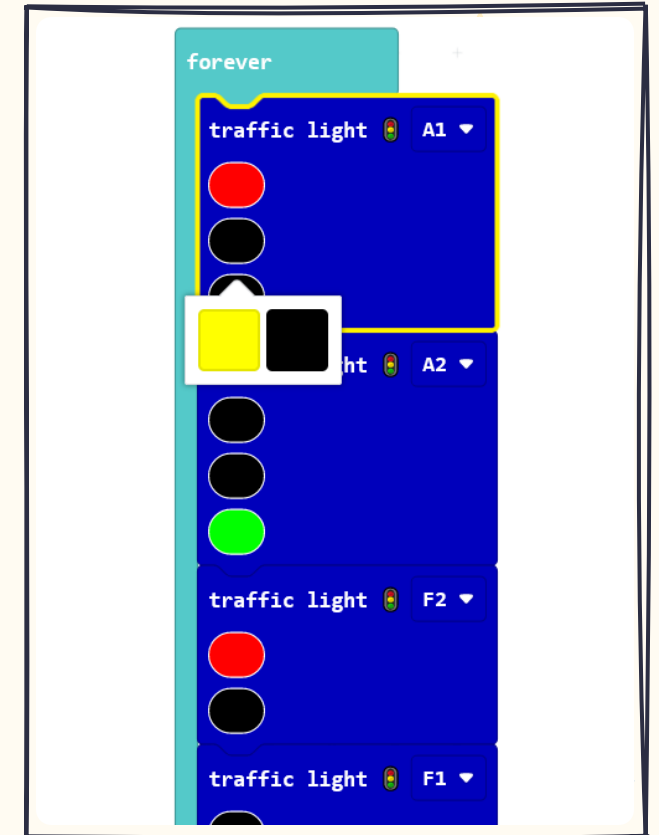
● turn OFF red light

3 easy steps

1. Snap the blocks **in order**
2. Press **Upload**
3. Watch your lights!

Magic word: WAIT

The wait block tells each light how many seconds to stay on.



Blocks look like this in our coding tool



Challenge time!

Can you do it?



Safe crossing: make green longer so children have time to cross.



Beep beep: add a buzzer that beeps when it is safe to walk.



Bug hunt: the lights are in the wrong order. Can you fix them?

What did we learn?



Red means stop, yellow means get ready, green means go



Instructions must be in the right order (a sequence)



A robot brain follows our blocks exactly

Tell your family tonight

Teach them what each traffic light colour means, and show them the green person when you cross together!



Next week: an Automatic Night Lamp that switches on in the dark! ✨

